Fake Nodes: Global polynomial interpolation on equispaced grids

Jaemin Oh

June 20, 2024 *

Can we use equispaced grids for global polynomial interpolation? One of the answers will be "NO", due to the Runge effect. Analyses based on the Lebesgue constant give us insights into why equispaced grids are bad, and why Chebyshev grids are good for global polynomial interpolation. However, we can use a simple trick to interpolate data on an equispaced grid without the Runge effect. In this post, we introduce Fake Nodes [1].

It's really simple. If we have $\{(x_i, y_i)|i = 0, 1, \ldots, N\}$ where $x_i \in [-1, 1]$, it forms a data on an equispaced grid. In this case, we define a map $S(x) = \cos(0.5\pi(x + 1))$ and let $s_i = S(x_i)$. S is a one-to-one map onto the set of corresponding Chebyshev points. Since we have $\{(s_i, y_i)|i = 0, 1, \ldots, N\}$ on a Chebyshev grid, Chebyshev polynomial interpolation gives quite an impressive result.

In fig. 1, $u(x) = 1/(1+x^2)$ is approximated with Chebyshev interpolant and FakeNode-Chebyshev interpolant with 31 points. The results are almost the same, indicating the effectiveness of Fake Nodes.

References

[1] S. De Marchi, F. Marchetti, E. Perracchione, and D. Poggiali, *Polynomial interpolation via mapped bases without resampling*, Journal of Computational and Applied Mathematics, 364 (2020), p. 112347.

*edit: June 20, 2024

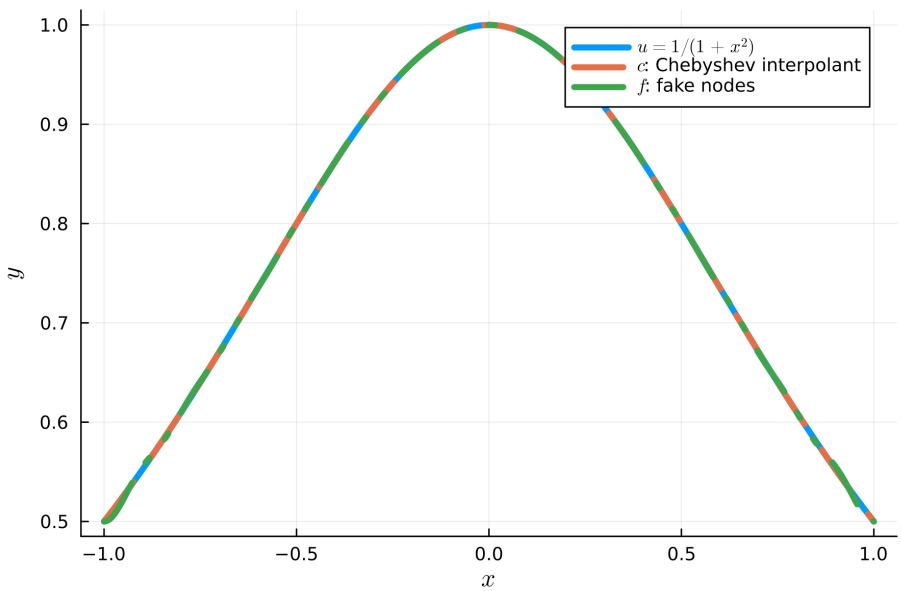

Figure 1: Approximation of $u(x) = 1/(1+x^2)$ with Chebyshev interpolant, and FakeNode-Chebyshev interpolant.